

Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)

On

An Intelligent Machine Learning Driven Framework for Predictive Risk Stratification of Diabetes Mellitus

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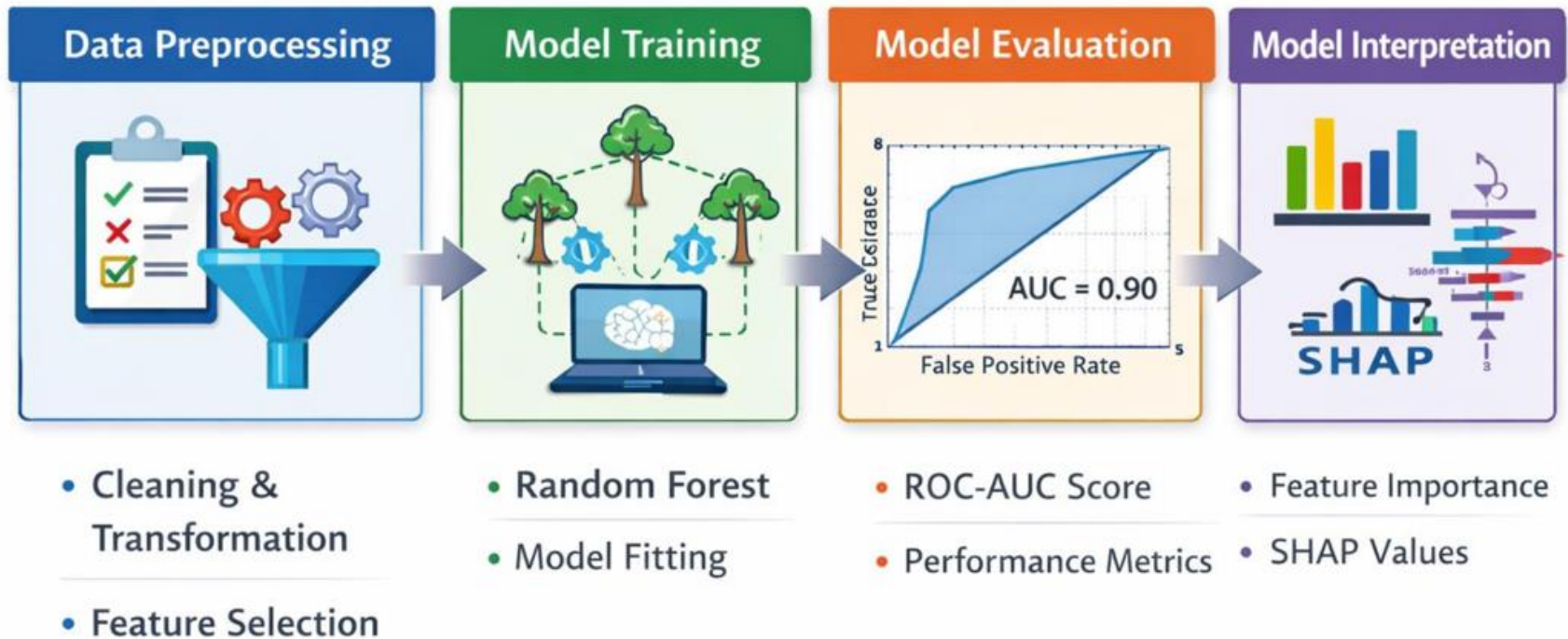
- Diabetes Mellitus is a rapidly growing global health issue
- Early detection is crucial to prevent serious complications
- Traditional diagnosis methods are time-consuming and manual
- Need for accurate and automated prediction system
- Machine Learning enables fast and accurate prediction
- Helps analyze medical data and identify hidden patterns
- Supports doctors in better decision-making
- This project focuses on building an intelligent and explainable prediction system

OBJECTIVE

- Develop an intelligent machine learning model for diabetes prediction
- Perform comparative analysis of multiple ML algorithms
- Improve prediction accuracy using advanced models like XGBoost
- Evaluate model performance using standard metrics
- Integrate Explainable AI techniques (SHAP & LIME)
- Identify key medical features influencing diabetes risk
- Build a system to support healthcare decision-making

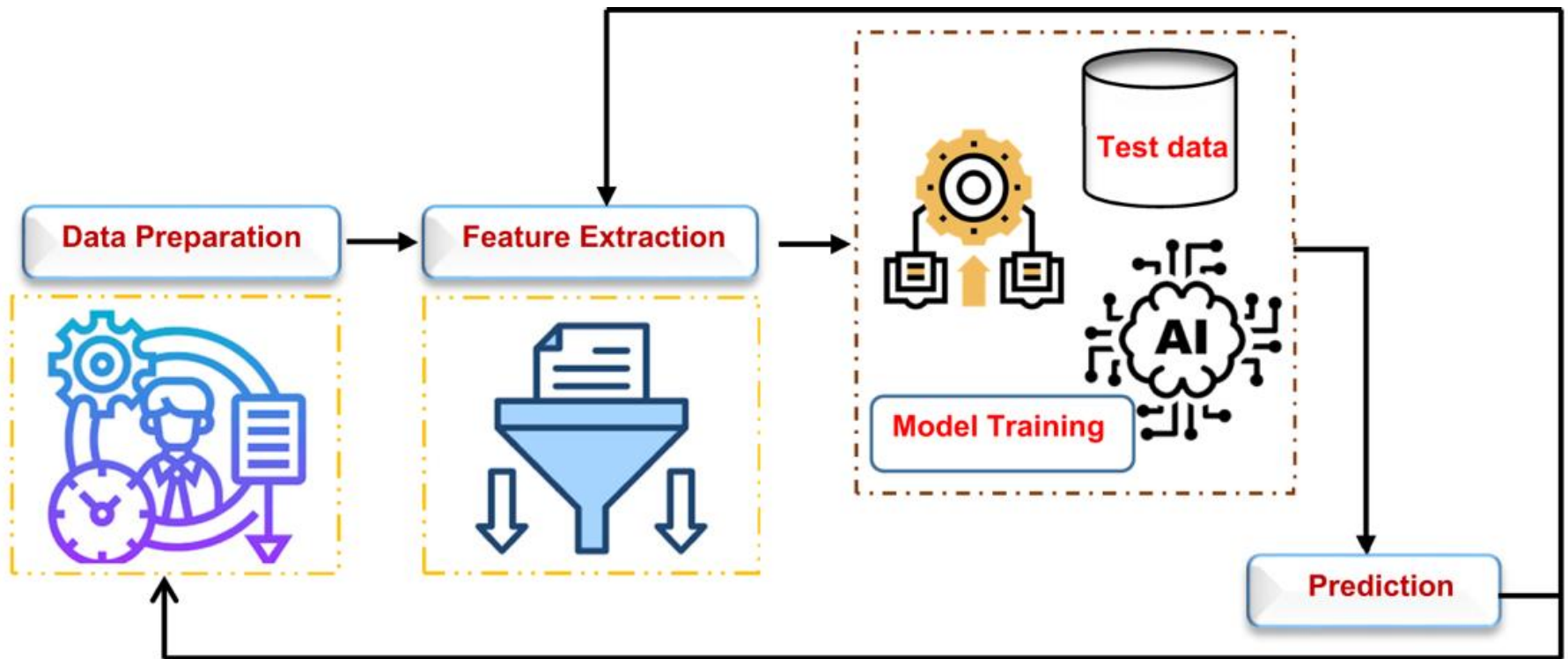


Machine Learning Pipeline for Healthcare Data



- Data preprocessing:
 1. Handling missing values
 2. Replacing invalid zero values
- Feature engineering and normalization
- Splitting dataset into training and testing sets
- Applying multiple machine learning algorithms
- Evaluating models using standard metrics
- Comparing model performance using graphs
- Applying Explainable AI techniques for interpretation

METHODOLOGY

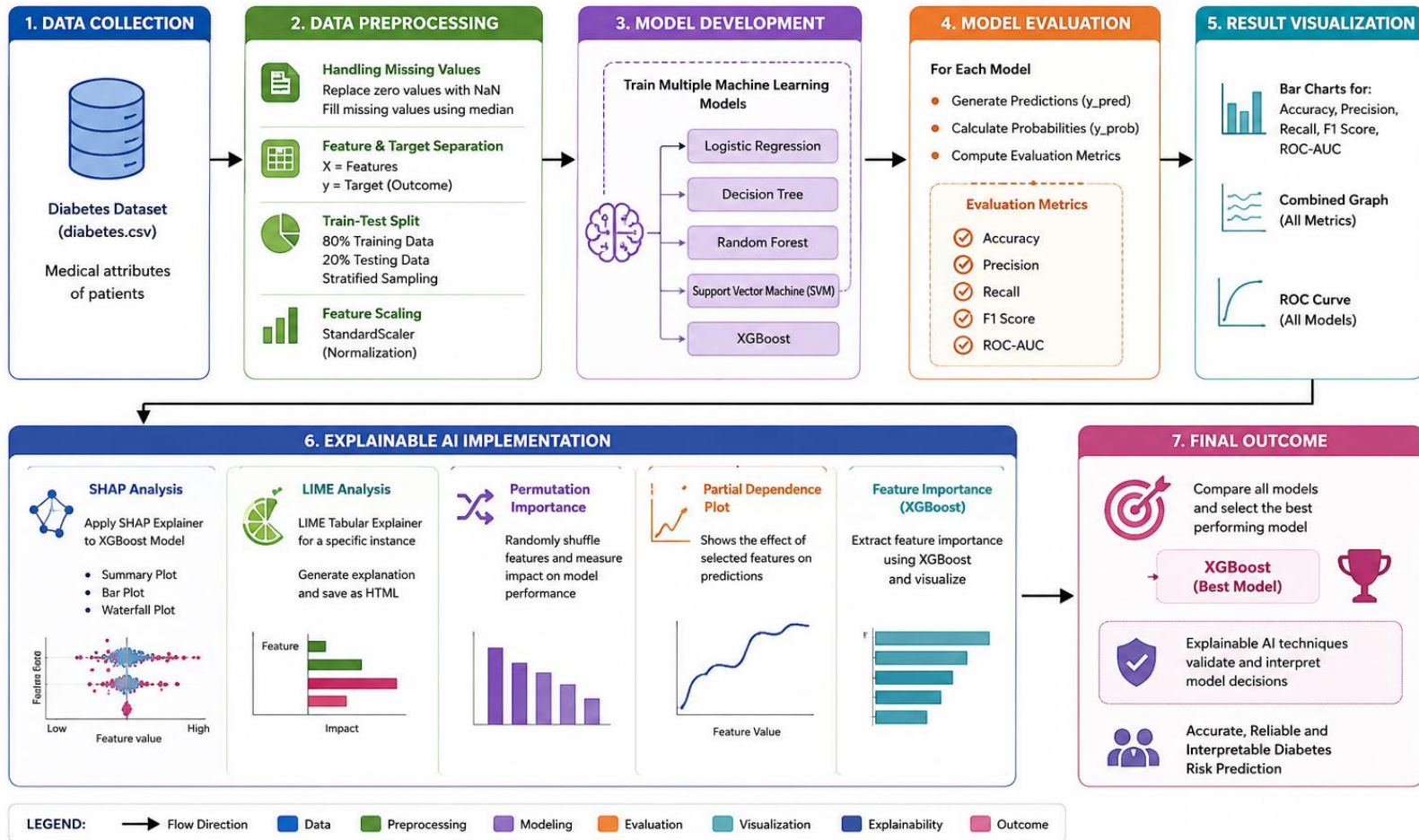


FLOWCHART



IMPLEMENTATION DIAGRAM

Machine Learning Driven Framework for Predictive Risk Stratification of Diabetes Mellitus



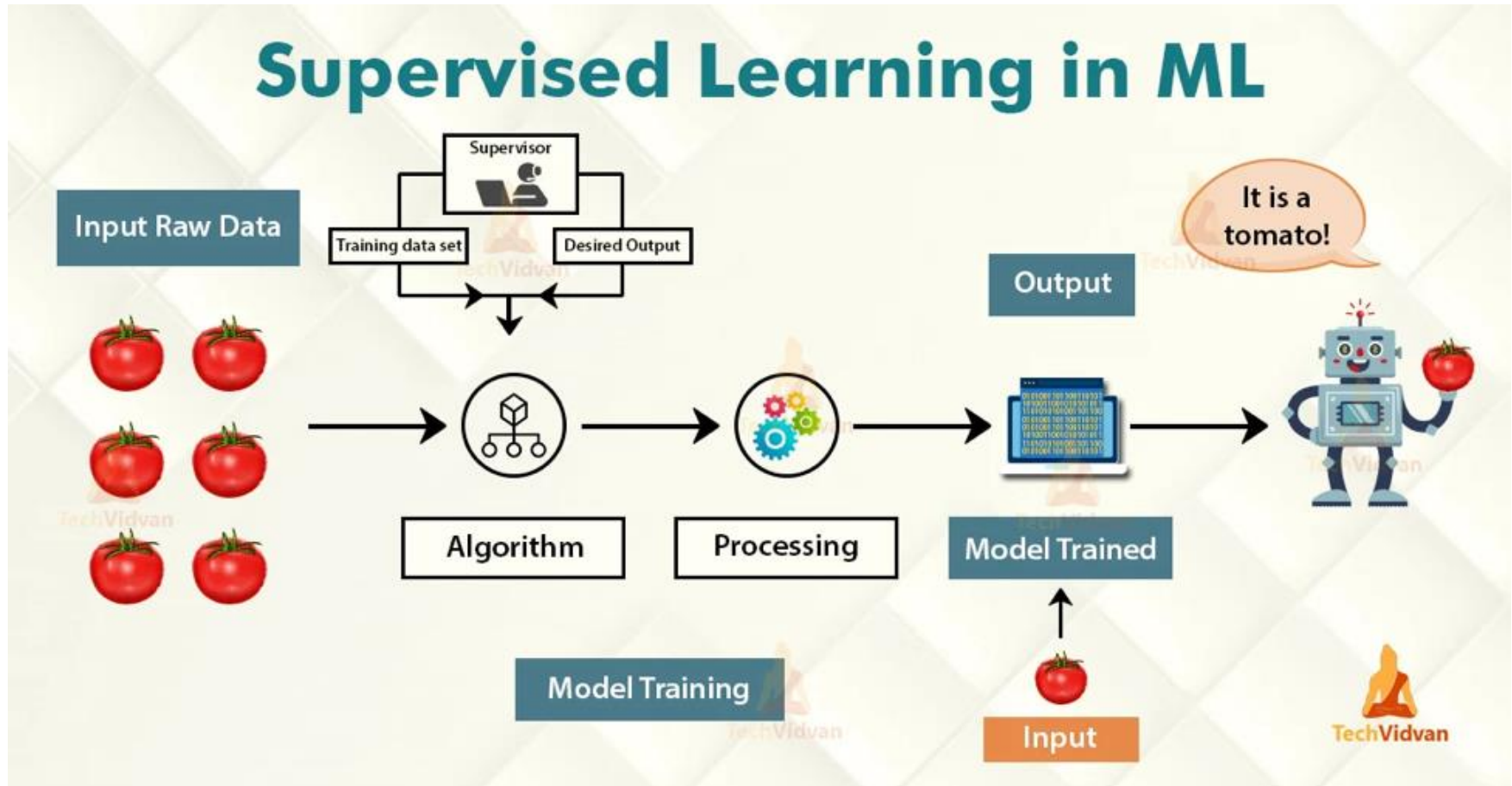
These are the **Supervised Learning** ML models used-

- Logistic Regression: Simple and interpretable baseline model
- Decision Tree: Easy to understand rule-based model
- Random Forest: Reduces overfitting using multiple trees
- Support Vector Machine: Works well with high-dimensional data
- XGBoost: Boosting algorithm with high efficiency

ML MODELS



Supervised Learning in ML



- **Accuracy:** overall correctness of the model
- **Precision:** correctness of positive predictions
- **Recall:** ability to detect diabetic cases
- **F1 Score:** balance between precision and recall
- **ROC-AUC:** model's ability to distinguish classes

RESULTS

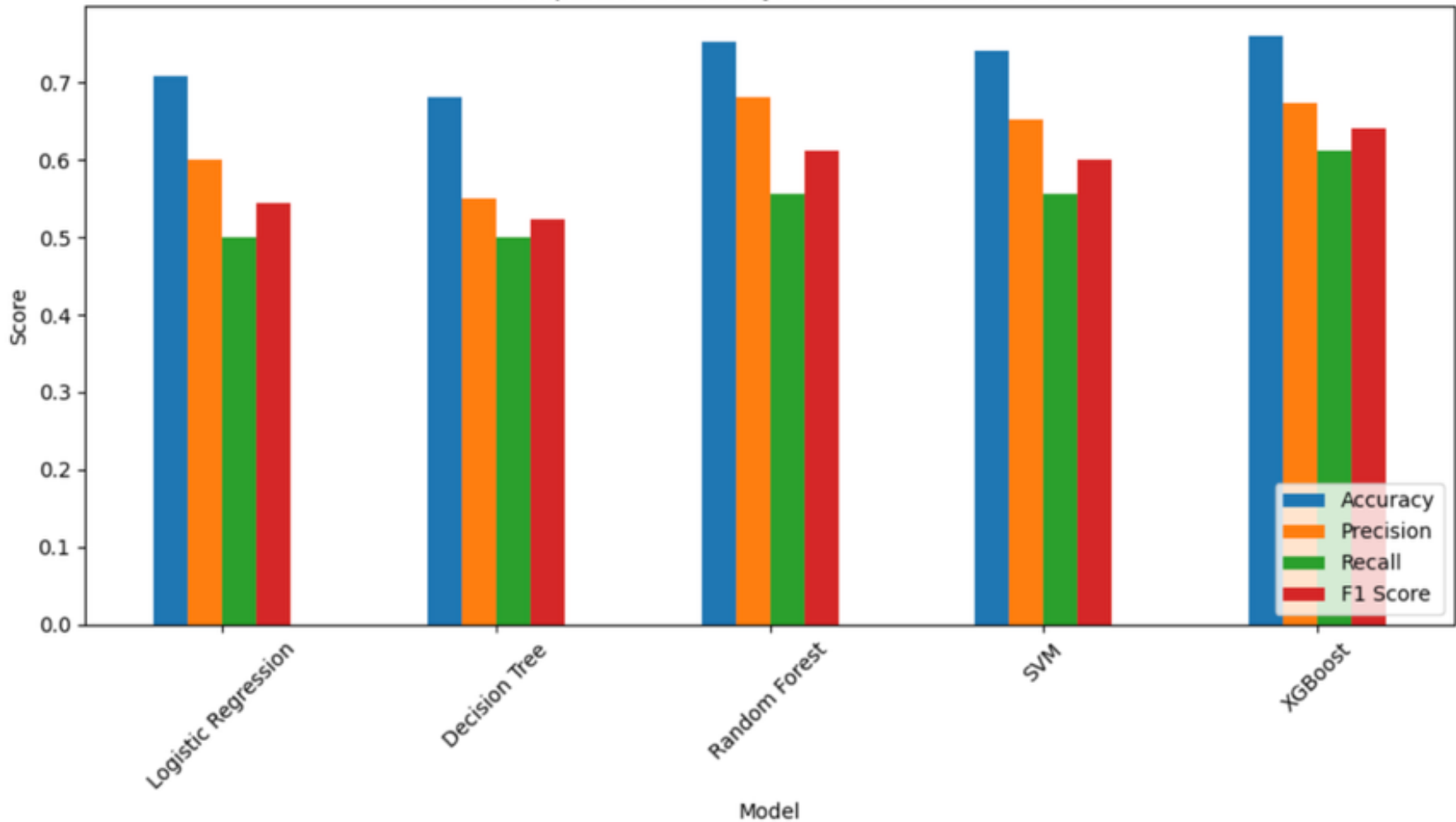
- XGBoost achieved highest accuracy and ROC-AUC
- Random Forest performed consistently well
- Logistic Regression showed stable baseline results
- SVM performed better after feature scaling
- Ensemble models outperformed single models
- Graphs clearly show performance differences
- ROC curve indicates strong classification ability

**Key Outcome -
XGBoost is the best-performing model**

RESULTS



Model Comparison (Accuracy, Precision, Recall, F1 Score)



- SHAP: Provides global feature importance
- Shows how each feature affects prediction
- LIME: Explains individual predictions
- Helps understand specific cases
- Improves transparency of ML models
- Builds trust among healthcare professionals
- Makes model decisions interpretable

CONCLUSION

- Machine Learning can effectively predict diabetes risk
- XGBoost achieved the best performance among all models
- Ensemble models provide higher accuracy
- Explainable AI improves transparency and trust
- Key features like Glucose and BMI strongly influence prediction
- System can assist in early detection of diabetes
- Supports better healthcare decision-making

- Integrate with real-time hospital databases
- Develop web or mobile application
- Use deep learning models for better accuracy
- Include lifestyle and genetic data
- Implement real-time monitoring system
- Deploy on cloud for scalability
- Continuously update model with new data



Thank You